

Accessisng the impact of Mental Health Influencers In Saudi Arabia

A Data Science analysis of youtube & tiktok content



3,700+ Videos Collected



2019-2025 Time Span



YouTube & TikTok APIs & data scraping

Introduction

Social media platforms such as YouTube and TikTok have become powerful spaces for discussing mental health in Saudi Arabia. Influencers play a key role in spreading awareness through educational content, personal experiences, and short-form videos.

Despite this growth, there is limited data-driven understanding of how this content affects audience engagement. This project analyzes mental health influencer content to uncover patterns and measure its impact.

Objectives

- How does engagement differ between YouTube and TikTok for mental health content?
- Which content features (views, likes, comments, video length) most influence engagement?
- How do influencer tiers (Nano, Micro, Macro) impact audience interaction?
- What mental health topics generate the highest engagement?
- Can machine learning models accurately predict engagement performance?

Data Collection

Data was collected from YouTube and TikTok using APIs and web scraping techniques.

- Total dataset: 3,713 videos
- YouTube: 3,336 videos
- TikTok: 377 videos
- Time span: 2019 – 2025
- Language: Arabic

Collected features include:

Views, Likes, Comments, Shares, Duration, Influencer Tier, and Content Type

Analysis

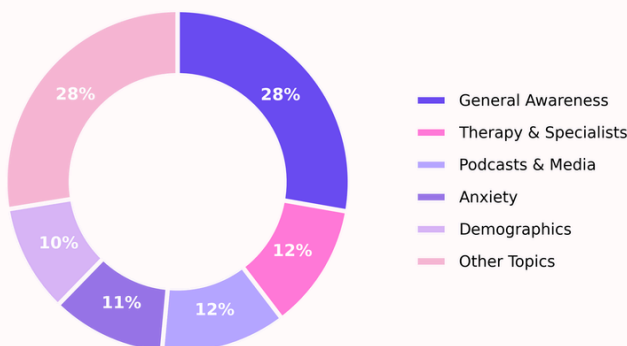
Exploratory Data Analysis revealed several key insights:

- TikTok shows higher engagement efficiency
- YouTube contains a larger volume of content
- Engagement distribution is highly skewed (few viral videos dominate)
- Strong correlation between views and likes

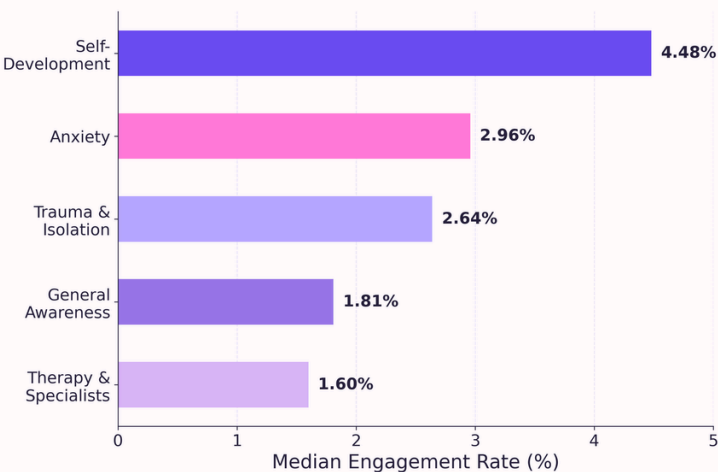
Top content topics:

- General Awareness
- Self-development
- Anxiety & Stress

Content Distribution by Topic



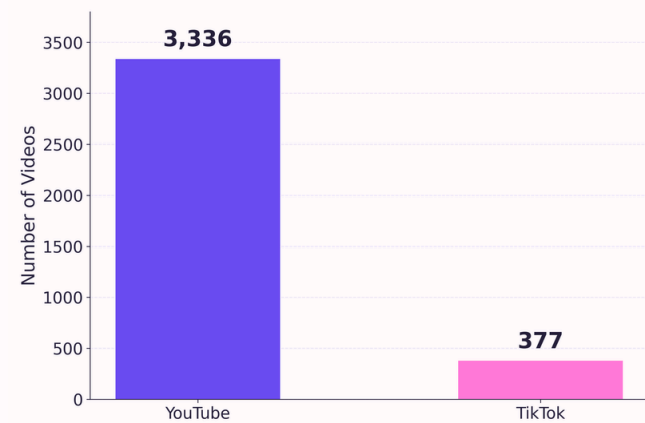
Median Engagement Rate by Topic



Correlation Between Engagement Metrics



Content Volume by Platform



Models & Findings

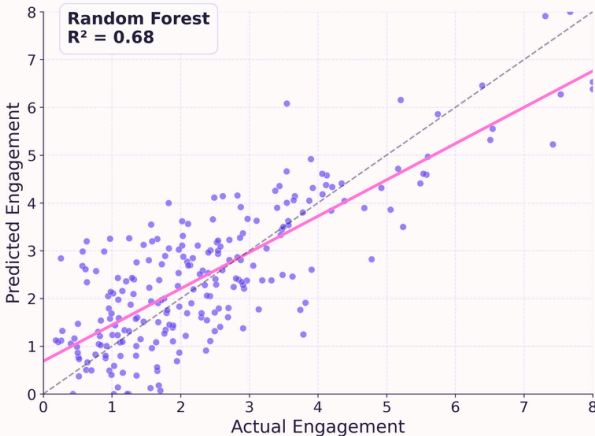
We applied several machine learning models to predict engagement performance, including Linear Regression,, and Random Forest.

The best-performing model was Random Forest ($R^2 = 0.68$), achieving the highest accuracy among all models.

Our analysis shows that platform type, content format, and video length significantly influence engagement.

Additionally, short-form content on TikTok demonstrated higher engagement efficiency compared to longer content on YouTube.

Actual vs Predicted Engagement



Conclusion

- Social media plays a major role in increasing mental health awareness
- TikTok is more effective for engagement, especially for short videos
- Micro and Macro influencers have the strongest impact
- Data-driven insights can improve content strategies and awareness campaigns

These findings help organizations and creators design more effective mental health content.



Towards a healthy, aware, and thriving society

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